Tampere University

BBT.HTI.501 Processing of Biosignals Assignment 2

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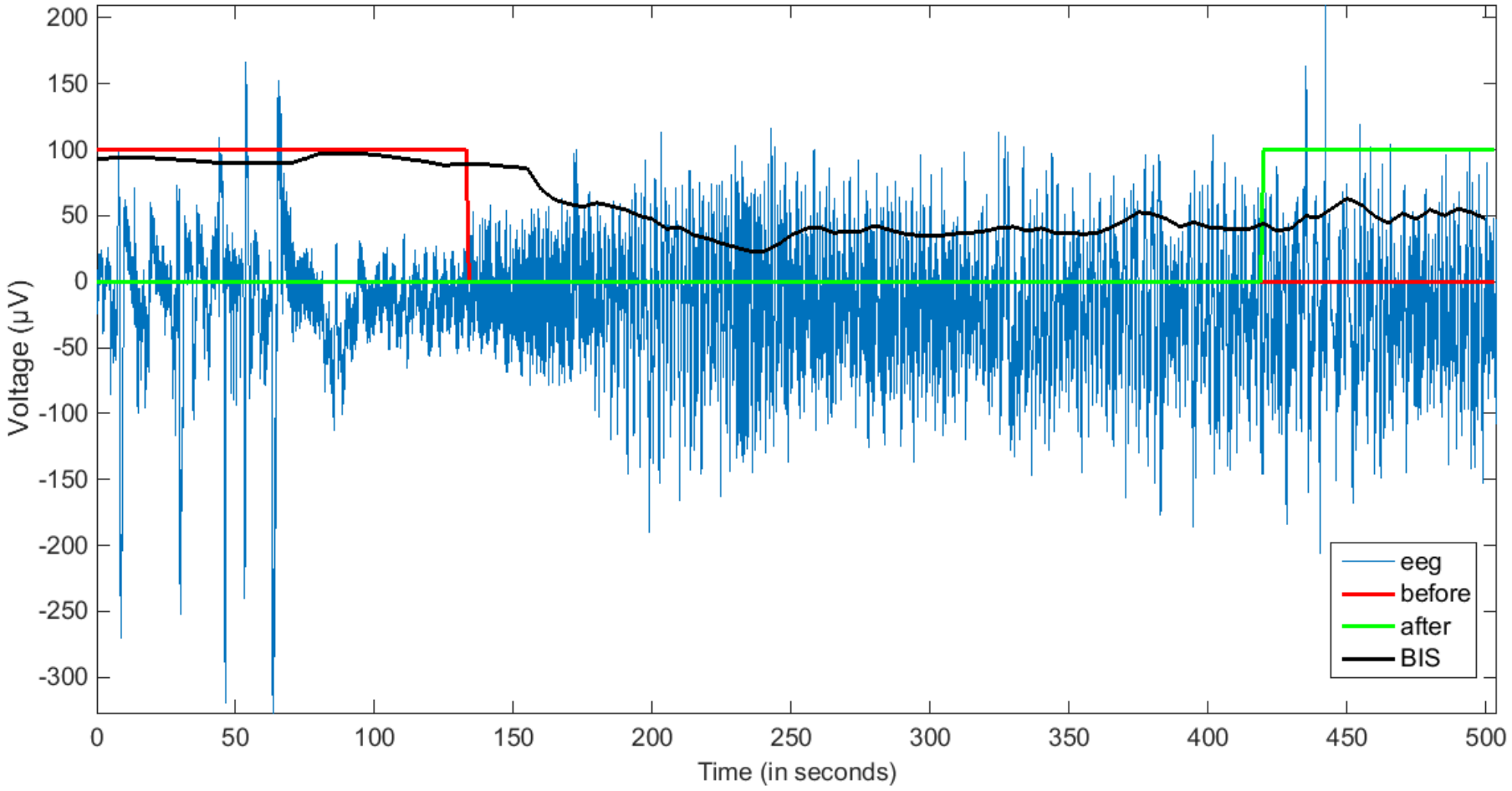
EEG & depth of anesthesia

- During anesthesia, EEG signal is more regular, i.e., less complex compared to when being awake
- The complexity/regularity of a signal can be assessed with entropy
  - In frequency domain: spectral entropy

→ The higher the spectral entropy of the EEG signal is, the more likely it is that the subject is awake

The data: EEG, anesthesia & BIS index



eeg = EEG signal
Before = has a value of 1 before the start of anesthesia, otherwise 0
After = has a value of 1 after the end on anesthesia, otherwise 0
BIS = BIS monitor values

FIR & IIR filters

- FIR filter needs higher order to meet the given frequency response than an IIR filter
- FIR filters have more ripple* in the stop- and passbands than IIR filters
- FIR filters are always stable, whereas IIR filters are not

*ripple = the fluctuation of voltage or current from the mean value

- You may either have a try at both filter types, or you can choose one and go with that (as you wish)

- **How do you calculate the sampling frequency from the dataset?**
  - Tip: Calculate the average time between two consecutive samples and calculate the fs from that. Sketch the idea first as a pseudo code for two samples and then expand the idea for rest of the samples.

Spectral entropy tips

- Remember to have a look at **entropy_eeg.m** function to see what is fed into your **spectral_entropy.m** function! Especially, have a look at rows
  - 33: “sf=…” and
  - 44: “H(cnt) = …”

- Carefully read through the equations in Spectral Entropy paper, it might be easy to sketch the algorithm on paper or as pseudo code before making the final code (at least if the algorithm idea seems difficult)
- **P** is the PSD of the EEG signal, that is the first input of your function. The other input is **nfactor**, check from **eeg_entropy.m** how to implement it in your code.

$$\sum_{f_i=f_1}^{f_2} P_n(f_i) = C_n \sum_{f_i=f_1}^{f_2} P(f_i) = 1 \qquad [3]$$

In the summation step, the spectral entropy corresponding to the frequency range $[f_1,f_2]$ is computed as a sum:

$$S[f_i, f_2] = \sum_{f_i=f_1}^{f_2} P_n(f_i) \log\left(\frac{1}{P_n(f_i)}\right) \qquad [4]$$

Thereafter, the entropy value is normalized to range between 1 (maximum irregularity) and 0 (complete regularity). The value is divided by the factor log ($N[f_1,f_2]$) where $N[f_1,f_2]$ is equal to the total number of frequency components in the range $[f_1,f_2]$:

$$S_N[f_1, f_2] = \frac{S[f_1, f_2]}{\log(N[f_1, f_2])} \qquad [5]$$

→ nfactor

Useful MATLAB commands

- Plotting the figures:
  - `hold on, legend, plot, xlabel, ylabel`
- Filter design & filtering:
  - `Butter, filtfilt, fir1, freqz`
- Note: when you calculate the sampling frequency, remember to **round up** the result!
- For FIR filter (fir1): $a = 1$ (For IIR filter (butter) you need to calculate both b and a parameters) → Check help and mathworks
- Scaling factor (sf) == nfactor
- Use the same logarithm as in entropy_eeg.m line 33!